DriveShield: FL-based Framework for Privacy Preserving and Jamming Attack Detection in V2X Communication Underlying 6G

Dhyey Thakkar*, Prince Tandel†, Aparna Kumari‡,
*†‡Department of Computer Science and Engineering Institute of Technology, Nirma University, Ahmedabad, India
Emails: *23bce540@nirmauni.ac.in, †23bce338@nirmauni.ac.in, ‡aparna.kumari@nirmauni.ac.in,

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***Abstract*—In this paper, we introduce *DriveShield*: a decentralized framework based on Federated Learning (FL) for real time jamming attack detection in VANETs under 6G communications. The local Feedforward Neural Network (FNN) trained on each client is aggregated by the popular Federated Averaging (FedAvg) approach and achieves a global accuracy around 83%, and reaches a peak of 87% across the training rounds. In order to preserve data privacy and integrity, the integration of blockchain and the InterPlanetary File System (IPFS) is used to make tamper resistant, secure model weight transfers. *DriveShield* is distinguished from such static dataset reliant approaches in the fact of their reliance on static datasets and real time data simulation of network conditions enhancing adaptability to dynamic network conditions. Performance evaluation shows that 6G-THz communication has better latency performance than 5g uRLLC, as well as better transaction scalability using IPFs. Robust detection capabilities with precision = 0.832—0.879 and F1-score = 0.822—0.880, over centralized baselines, are confirmed via metrics.**

***Index Terms*—Federated Learning, Jamming Detection, Neural Networks, DriveShield, Data Security & Privacy, Federated Averaging, Blockchain, VANET, 6G, IPFS.**

I. Introduction

Currently, the rising need for Vehicular Ad Hoc Networks (VANETs) in today's transportation has changed road safety, traffic management, and time-sensing vehicular communication. Vehicles interacting with infrastructure with no obstacles produce improved driving performance and decreased roadway safety threats through VANETs. Open wireless channels expose VANET systems to potential jamming attacks which produce system breakdowns and delay emergencies together with risking road safety. Current jamming detection approaches typically employ centralized approaches, which have various disadvantages such as limited scalability, privacy risks, as well as weak presence to a few single points of failure [1].These systems need to communicate with a server causing delays and making them more vulnerable, to attacks, from outsiders.The use of machine learning methods has been adopted to identify jamming attacks by examining network interruptions [2]. However, such methods achieve this goal at the cost of large quantities of vehicle data, which causes privacy worries and ramps up processing loads. To tackle these challenges, there exists an urgent requirement for a decentralized and privacy-sensitive strategy that can promptly identify jamming attacks without losing any security or efficiency.

FL has been used over in lots of studies for privacy-preserving and decentralized security solutions. FL supports cooperative model training without revealing the original data, however, its use in such VANETs' Jamming Attack detection remains underexplored [3], [4]. To handle this issue, we present DriveShield, a decentralized framework that utilizes FL for effective detection of jamming attacks while maintaining the privacy of each vehicle.

To enhance the security of DriveShield further blockchain technology is utilized to introduce a level of protection, transparency and reliability. [5], [6] Utilizing the blockchain's distributed and immutable property, all updates of the model and transactions are recorded securely to prevent unauthorized changes. Every update is time-stamped and stored on an immutable ledger, so no data can be changed or tampered with. Moreover, model upgrades are preserved utilizing an off-chain method such as InterPlanetary File System (IPFS), which boosts security as well as performance. With the integration of blockchain and Federated Learning, DriveShield ensures data privacy as well as prevents the presence of persistence of the implemented design on the jamming attack detection integrity verified in VANETs.

A. Novelty

The novelty of *DriveShield* is that it presents an FL-based decentralised framework that performs secure real-time jamming detection in vehicular communication networks for VANETs and V2X systems. Unlike traditional centralised models, *DriveShield* trains a FNN individually on each client and uses a FedAvg strategy to modify the weights considering the insights from all clients. Real-time data simulation runs to reflect changing network conditions is one of its major strengths because other research fails to include this functionality. *DriveShield* depends on Blockchain and IPFS systems to create tamper-proof weight exchange which solves the security problems that exist during FL-based training. *DriveShield* implements a system that operates under low-latency V2X conditions to connect fast jamming detection methods with privacy-focused vehicle network intelligence.

B. Research Contributions

Here is the research contribution of the paper:

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- Designing *DriveShield* Framework: we introduce *DriveShield* is a real-time jamming detection framework based on FL which enables real-time jamming detection on VANET.
- Implementing Privacy-Preserving Techniques: Client Specific FNN Training with FedAvg Aggregation. Specifically, DriveShield has an FNN on each client and FedAvg in the aggregation of model updates between vehicles. Hence, the approach facilitates collaborative learning with data privacy and a fast convergence of the global model.
- Evaluation of the proposed *DriveShield*: The proposed framework is evaluated based on various parameters like accuracy, precision, F1-score and proved its effectiveness compared to the existing approach.

C. Article Layout

Rest of the paper is organized as follows. Section II explores the existing literture. Section III describes the system model and problem formulation. Section IV discusses the privacy constraints. Section V results and performance analysis of the proposed framework. Lastly Section VI concludes the paper.

II. Literature Review

Dimitrios et al 2018 [1], introduce an unsupervised machine-learning method for sensing RF jamming attacks in vehicular networks (VANETs). Key contributions entail the use of a variant speed of vehicle, which is thought to describe speed variations between jammer and receiver, along with other classical parameters such as the packet delivery ratio (PDR), signal interference-to-noise ratio (SINR), and the received signal strength indicator (RSSI). Besides kosmanos et al 2021 [2], used the supervised learning-based RF jamming detection scheme for VANETs using K-Nearest Neighbors (KNN) and Random Forest (RaFo). This model is evaluated under three attack scenarios with good detection accuracy, especially when the training keeps high vehicle speeds of above 15m/s. The detection system is centralized and uses TCA for classification results. Punal et al 2015 [7], analyze the effects of RF jamming on VANET communications based on the standard 802.11p, showing interference that is constant and periodic in nature. These can severely disrupt car-to-car communications which proves an urgent need for protocols able to protect against jamming.

Another work by Mowla et al. 2019 [8], proposes a federated learning-based model to detect jamming attacks in Flying Ad-hoc Networks (FANETs). It focuses on using a federated learning model in training datasets for CRAWDAD and ns-3 simulated FANET architectures. It presents a novel security architecture to improve federated learning in jamming attack detection. By Kim et al 2024 [9], studied the impact of jamming attacks on blockchain-based federated learning (BCFL) operating in wireless networks. The analysis performed through simulations evaluated both the chance of attack success and the levels of miner involvement. which shows that Security measures need to establish robust defenses to protect FL systems. Besides Ali et al 2023 [10] explored implementation of Blockchain together with Federated Learning (FL) creates secure Industrial IoT (IIoT) networks through decentralized intrusion detection procedures which respect privacy. where FL keeps data stored locally to protect it from cyber attacks but Blockchain provides secure tamper-proof logs for device trust. By Huang et al 2024 [11] This survey document assesses modern federated learning progress through studies on both generalization and robustness along with fairness aspects [12], [13]. Sameera et al 2024 [14], The research evaluates Blockchain-enabled Federated Learning (BCFL) as an approach to conducting decentralized model training with privacy and security features [15].

III. System Model and Problem Formulation

This section outlines the system model and problem formulation for federated learning in credit risk assessment.

A. System Model

The Federated Learning (FL) system model is designed for securing Vehicular Ad Hoc Networks(VANETs) is shown in Fig. 1,works without a constant data exchange with a central server. Data are gathered from various sources, including roadside units, cellular towers, jammers, and various vehicles, and are preprocessed in a centralized database. After this point, a Feedforward Neural Network (FNN) is trained with multiple local client models independently. We have used FNN model at client level keeping in mind the low computational capablity of vehicular nodes. Instead of sharing raw data, these clients send only the model weights to the server via a secure blockchain and IPFS framework, thus lowering the communication overhead on clients. Subsequently, the server receives the weights and builds its global model via an RPC server using the FedAvg strategy to ensure data safety and privacy. Our model consists of four layers and The federated learning process is iterative:

1) The Data Acquisition and Sensing Layer: Collects raw data from vehicles, roadside units, cellular towers, and jammers, storing it in a dataset for further processing.
2) The Edge Processing and Machine Learning Layer : Preprocesses data and trains client-side FNN models independently at the edge.
3) The Secure Communication and Blockchain Layer: Ensures secure transmission of model weights using blockchain and IPFS, maintaining data integrity and privacy.
4) The Federated Learning and Aggregation Layer: Aggregates local model weights using an RPC server and the FedAvg strategy to construct a global model. and the updated global model is shared back with all institutions to continue local training in the next iteration.
5) The process repeats until the global model converges or a performance threshold is reached.

B. Problem Formulation

DriveShield is a framework that can detect jamming attacks in vehicular communication networks while preserving privacy and security using FL. The key interest is the minimization

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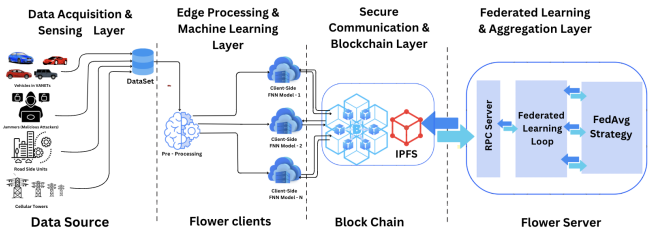

Fig. 1: *DriveShield* Framework: System Model

of the loss function and the guarantee of the effective model generalization over all clients. The algorithm 1 shows the working.

The problem can be formulated as:

$$\min_{w} \frac{1}{N} \sum_{i=1}^{N} L(f(x_i; w), y_i) + \lambda R(w) \tag{1}$$

where:

- N representing the number of vehicles (clients) participating in FL.
- x_i and y_i denotes the input features and corresponding jamming labels for the client i.
- L is the classification loss function (e.g., cross-entropy).
- $f(x_i; w)$ is the FNN model with parameters w.
- $R(w)$ is the regularization function to enhance generalization.
- λ is a weight factor controlling regularization strength.

1) Feedforward Neural Network (FNN) with ReLU: Each client locally use an **FNN** with **ReLU activation** in hidden layers:

$$h^l = \text{ReLU}(W^l h^{l-1} + b^l) \tag{2}$$

where:

- W^l, b^l are the weight matrix and bias vector for layer l.
- h^l is the activation at layer l.
- **ReLU activation function:** $\text{ReLU}(x) = \max(0, x)$.

C. Loss Function: Categorical Cross-Entropy

Since *DriveShield* deals with **multi-class jamming detection**, it uses **Categorical Cross-Entropy Loss**:

$$L(y, \hat{y}) = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{3}$$

where:

- y_i is the **one-hot encoded** ground truth label.
- $\hat{y}_i$ is the Softmax output.

1) Federated Learning with FedAvg for Multi-Class Classification: Each client **trains locally** using a subset of the dataset (either **Dataset 1** or **Dataset 2**). The **FL server** aggregates models using **Federated Averaging (FedAvg)**:

$$W^{t+1} = \sum_{i=1}^{N} \frac{|D_i|}{\sum_{j=1}^{N} |D_j|} W_i^t \tag{4}$$

$$b^{t+1} = \sum_{i=1}^{N} \frac{|D_i|}{\sum_{j=1}^{N} |D_j|} b_i^t \tag{5}$$

where:

- W^{t+1}, b^{t+1} are the updated **global model parameters**.
- W_i^t, b_i^t are the **local model parameters** from client i.
- $|D_i|$ is the number of **training samples** at client i.

Algorithm 1 *DriveShield*: Federated Learning-Based Jamming Detection

procedure DRIVESHIELD($C, \{\mathcal{D}_c\}_{c=1}^{C}, E, T$)
  Input: C - Clients, $\mathcal{D}_c$ - Local datasets, E - Local epochs, T - Communication rounds
  Output: Trained global FNN model for jamming detection
  Initialize global model W^0
  for $t \leftarrow 0$ to $T-1$ **do** ▷ Federated Learning Rounds
    for each client $c \in C$ **in parallel do**
      Load dataset $\mathcal{D}_c$, normalize features
      Train FNN locally for E epochs using Adam optimizer
      Compute loss:

$$L(y, \hat{y}) = -\sum_{i=1}^{3} y_i \log(\hat{y}_i)$$

      Compute evaluation metrics (Accuracy, Precision, Recall, F1-score)
      Send updated weights W_c^{t+1} to server
    end for
    Server Aggregation: Update global model using FedAvg:

$$W^{t+1} = \sum_{c=1}^{C} \frac{|\mathcal{D}_c|}{\sum_{j=1}^{C} |\mathcal{D}_j|} W_c^t$$

    Distribute updated model W^{t+1} to clients
  end for
  Jamming Detection: Predict scenario using Softmax:

$$\hat{y} = \arg\max_i (\hat{y}_i)$$

  return Final global model W^T
end procedure

IV. Privacy Constraints

The *DriveShield* framework is a tool that makes sure that vehicles do not transmit raw communication data, which is a violation of people's privacy and the security of vehicles. Instead, each car invents the best local model on its dataset and only transmits encrypted model updates (like weights) to the federated learning server. The following procedures are employed to maintain privacy and prevent adversarial attacks during model aggregation:

- **Secure Aggregation:** The FL server aggregates model updates from multiple clients using *FedAvg* while preventing individual model parameters from getting exposed.
- **Blockchain-based Integrity Verification:** Model updates are applied to the blockchain in an IPFS hash and can be kept tamper-proof during the weight exchanges while making the unauthorized modifications difficult and ineffective.

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V. Result Evaluation

The results evaluation section assesses the performance of the proposed framework by comparing its accuracy, precision, recall, and F1 score against traditional centralized models.

A. Experimental Setup and Dataset Description

1) Experimental Setup: DriveShield is implemented on system running macOS Sequoia 15.2 with 16GB RAM and an Apple M3 chip. This configuration ensures efficient execution of the framework, leveraging the advanced processing capabilities of the M3 chip for optimal performance.

2) Dataset Description: The RF jamming dataset [16] for vehicular wireless networks which is a whole dataset designed for evaluating certain types of jamming detection algorithms in VANETs. It contains real measurements of wireless communication features including Signal-to-Noise Ratio (SNR), Received Signal Strength Indicator (RSSI), Packet Delivery Ratio (PDR), Speed, and Relative Speed.

The dataset is set out in the following three scenarios:
Scenario 1 (No Attack), where only background interference is present;
Scenario 2 (Reactive Jammer Attack), in which a jammer transmits signals only now and then while pursuing a target.
Scenario 3 (Constant Jammer Attack), which is when the jammer continuously transmits at maximum power.
Also, the data set is divided in two subsets based on different relative speed thresholds (15m/s and 25m/s) to reflect various mobility conditions. This dataset will prove valuable to the development of machine learning-based jamming detection models in V2X communication systems as it will be both the source of the real life data and the basis of the requirements for performance evaluation.

B. Performance Metrics

We analyze the performance of the *DriveShield* framework using typical classification measures, including:

- **Accuracy:** The percentage of successfully identified instances (default or non-default).
- **Precision:** The proportion of true positive predictions compared to all positive forecasts.
- **F1-Score:** F1-Score is the harmonic mean of precision and recall.
- **Scalability:** It is the amount of time required to process given number of transactions in IPFS.

The goal is to achieve a balance of model performance (accuracy, precision, and F1-Score) while maintaining privacy and regulatory compliance.

C. Result and Analysis

Experiments were conducted with 5G-uRLLC and 6G-THz-Comm. The results, visualized in the Fig. 2b, shows significant decrease in network latency in 6G-THz-Communication compared to high latency in 5G-uRLLC.

Table I and Fig. 2a shows the FedAvg algorithm's performance across 15 training rounds, with the essential metrics: accuracy, precision, and F1-Score. The accuracy fluctuates slightly in the beginning rounds and remains pretty steady in the final rounds, ranging between 0.826 and 0.869, suggesting the learning in the model's predictions.

As observed in the table I, it can be seen how precision is more or less on the same track as accuracy, as it remains at the same level between 0.832 and 0.879. At Round 4, the model reached its highest precision of 0.8794, which means the model obtained its best positive predictive value at the beginning of the algorithm. Nevertheless, the overall precision constancy from the lenient degrees to the stricter ones makes certain the existence of a healthy model, that is, a model that correctly identifies every positive instance every time.

The F1-Score, which is calculated by combining precision and recall, also ranges from 0.8238 to 0.869, confirming that the model is good at reducing false positives and negatives during training. The highest F1-Score at a rate of 0.869) is observed in Round 5, thus also consistent with the trend in accuracy and precision.

Experiments were conducted with 5G-uRLLC and 6G-THz-Comm. The results, visualized in Fig. 2b, which shows a significant decrease in network latency in 6G-THz-Communication compared to high latency in 5G-uRLLC.

Fig. 2c shows the comparison of transaction scalability of IPFS with existing blockchain technology currently used [17]. It can be seen that the scalability is significantly improving with reducing the Transaction Processing Time.

On the whole, the results reveal that the FedAvg algorithm works well and draws from the distributed data in a pattern noticed in a few rounds. The resignation of accuracy, precision and F1-Score by a few percentages may suggest that the model keeps the generalization level stable round by round, without any noticeable overfitting or distortion of performance.

TABLE I: Results of Model Performance Across Rounds

Round	Accuracy	Precision	F1 Score
1	0.8423	0.874	0.8352
2	0.8655	0.8719	0.8654
3	0.869	0.8744	0.8685
4	0.8695	0.8794	0.8686
5	0.8697	0.8769	0.869
6	0.8673	0.8787	0.8658
7	0.8662	0.8779	0.8645
8	0.8593	0.8703	0.8576
9	0.845	0.8526	0.8429
10	0.8342	0.841	0.8318
11	0.8295	0.8357	0.8271
12	0.8277	0.8335	0.8252
13	0.8265	0.8322	0.824
14	0.8263	0.832	0.8238
15	0.8267	0.8323	0.8241

Fig. 2d shows the comparison of proposed framework with the frameworks proposed in existing work [2] [18]. It can be seen the model clearly outperforms both the models while preserving privacy of the clients using FL and Blockchain. However, there are other models achieving more accuracy on the cost of providing low privacy using centralized model training.

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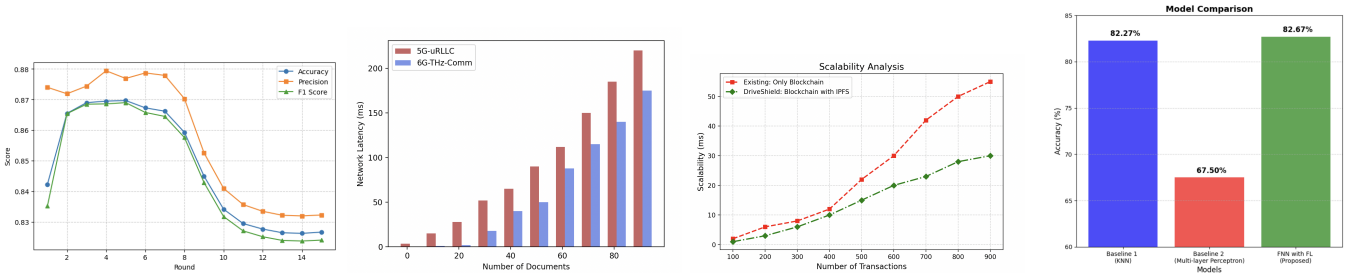

(a) (b) (c) (d)

Fig. 2: (a) Model Performance over Rounds, (b) Network Comparison result, (c) Blockchain Comparison result, and (d) Model Comparison with Baselines.

VI. Conclusion

In this paper, *DriveShield* is presented, which is built on FL with the FedAvg algorithm to enable the model to be trained in a decentralized fashion while preserving privacy and security in VANETs. The platform integrates Blockchain-based IPFS for secure and tamper-resistant model updates, making unauthorized access risk minimal. The platform being decentralized is a fitting solution for V2X jamming detection that happens in real-time such that different clients can mutually elevate the detection accuracy but with no necessity to share the raw data. Our model has succeeded by having an accuracy of 82.67%, a fact that proves the effectiveness of modern intelligent transportation systems.

In future, advanced FL algorithms that include FedProx and FedMA will be the next step in the development. It can also incorporate the scaling improvement of large-scale vehicular networks so that they always operate in high-density mobility and in dense traffic condition without any interruption. On-the-fly adaptation of FL techniques such as personalized FL or meta-learning is worth to check out for the authors to deal with the dynamic jamming patterns.

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